CHEMISTRY 101 · RECITATION HOMEWORK

Homework 2: Acids & Bases

Name: ______________________________ Date: ____________________

Instructions: About 20 minutes. Part C is the heart of it: read each blood gas like a detective — two clues, one story. The home lab at the end is optional and fun. Total: **25 points**.

Part A — Sort the kitchen

1. For each substance, write whether it is an acid, a base, or neutral — and whether it would taste sour, feel slippery, or neither. (1 point per row.) (6 pts)

Substance	Acid, base, or neutral?	Sour, slippery, or neither?
Lemon juice		
Baking soda in water		
Pure water		
Vinegar		
Hand soap		
Milk		

Part B — The ten-times ruler

2. A drink is pH 6. Another is pH 4. How many times more acidic is the pH 4 drink? Show how you got it. (2 pts)

3. Which is more acidic: pH 3 or pH 5? (1 pt)

4. Baking soda (a base) is stirred into vinegar (an acid) and it fizzes. What is happening? (1 pt)

- [ ] **A.** The acid and base are neutralizing each other, and the fizz is CO_2 escaping.
- [ ] **B.** The base is turning into a stronger acid, and the fizz is oxygen escaping.
- [ ] **C.** The vinegar is boiling because the baking soda made it hot.
- [ ] **D.** The two are mixing into soap, and the fizz is soap bubbles.

Part C — Blood gas detective

Normal values, about: pH 7.35-7.45; CO_2 35-45; bicarbonate 22-26. Remember the two stories: pH low + CO_2 high = the lungs aren't clearing acid (a breathing problem). pH low + CO_2 normal + bicarbonate low = acid is coming from starving cells (find the cause). pH high + CO_2 low = breathing too much.

5. Four babies, four blood gases. For each, write what the problem is and what the team should do. (2 points per baby: 1 for the problem, 1 for the plan.) (8 pts)

Baby	pH	CO_2	Bicarb	What's the problem?	What should the team do?
A	7.40	40	24		
B	7.20	70	25		
C	7.18	38	14		
D	7.52	25	24		

Part D — Flashback to Week 1 Chemistry

6. Carbonic acid is H_2CO_3 and baking soda (sodium bicarbonate) is $NaHCO_3$. How many atoms are in each molecule? (Na is sodium — one atom.) (2 pts)

Part E — Flashback to NICU Tutor

7. In the NICU, caffeine is a medicine. What is it for — and using today's chemistry, explain how it helps keep a preemie's blood pH steady. (2 pts)

Part F — Think like a doctor

8. A preemie has a 25-second apnea spell with a brady. The nurse rubs her back and she restarts. Twenty minutes later a blood gas shows pH 7.24 and CO_2 65. Explain what happened in her blood during the spell, and say two things the team might do next. (3 pts)

HOME LAB (optional, not graded): The red-cabbage pH tester

You need: a few red-cabbage leaves, hot water (grown-up helps), a jar, small cups, and things to test: lemon juice, vinegar, milk, plain water, baking-soda water, soapy water

1. Chop the cabbage, cover it with hot water, wait 10 minutes, and pour off the purple juice. That's your indicator.
2. Put a little of each test liquid in its own cup and add a spoonful of cabbage juice to each.
3. Watch the colors: pink or red means acid, blue or green means base, purple means neutral.
4. Line the cups up from reddest to greenest. That's a pH ruler. Write the order below, then try mixing the lemon cup into the baking-soda cup and watch the color (and the fizz).

What I saw:

Done! Professor's initials: ________

Score: __________ / 25 points

Professor's comment: ____________________________________